

Multiple Choice (10 marks)

1. The digest created by a hash function is normally called a
 - (a) Modication detection code (MDC)
 - (b) Modify authentication connection
 - (c) Message authentication control
 - (d) Message authentication cipher
2. Which of the following styles of fuzzer is more likely to explore paths covering every line of code in the following program?
 - (a) Generational
 - (b) Blackbox
 - (c) Whitebox
 - (d) Mutation-based
3. The _____ is anything which your search engine cannot search.
 - (a) Haunted web
 - (b) World Wide Web
 - (c) Surface web
 - (d) Deep Web
4. SHA-1 has a message digest of
 - (a) 160 bits
 - (b) 512 bits
 - (c) 628 bits
 - (d) 820 bits
5. Which of the following is true of mutation-based fuzzing?
 - (a) It generates each different input by modifying a prior input
 - (b) It works by making small mutations to the target program to induce faults
 - (c) Each input is mutation that follows a given grammar
 - (d) It only makes sense for file-based fuzzing, not network-based fuzzing

True / False (10 marks)

Answer True or False

6. True or False: A proxy firewall filters traffic at the application layer, inspecting the data for security threats and enforcing policies.
7. True or False: In message confidentiality, the transmitted message should only be comprehensible to the intended receiver.
8. True or False: BurpSuite is primarily designed for analyzing wireless traffic and is commonly used for network sniffing.
9. True or False: Message nonrepudiation ensures that a sender cannot deny having sent a specific message.
10. True or False: Message confidentiality focuses on preventing unauthorized access to data rather than ensuring its exact delivery.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a python function to find the sum of absolute differences in all pairs of the given array.
12. Write a function to add two integers. however, if the sum is between the given range it will return 20.

End of Paper